Reduced Units

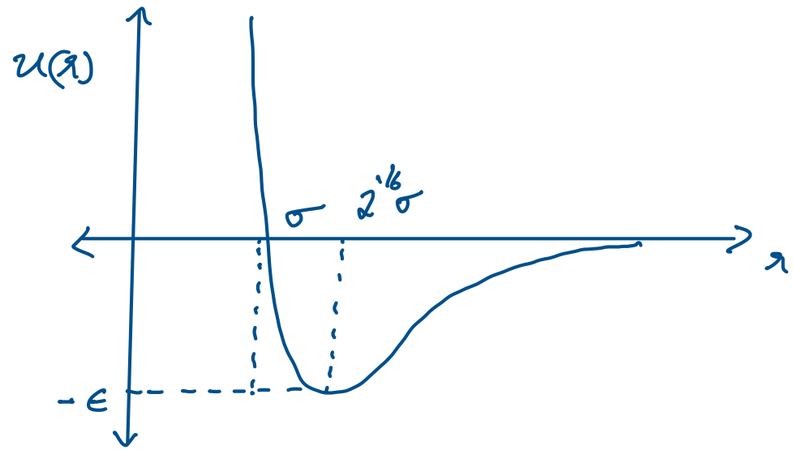
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It is important to choose appropriate units such that the numerical values of various quantities are neither too high or too low.

For Lennard-Jones systems, one can choose the following set of units such that our results can be scaled for any value of 'σ' and 'ϵ'.

Lennard-Jones potential

$$u(r) = 4\epsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right]$$



The property of any fluid will depend on the value of σ and ϵ

However, it is possible to scale the potential such that the properties of all materials can be obtained from a single simulation

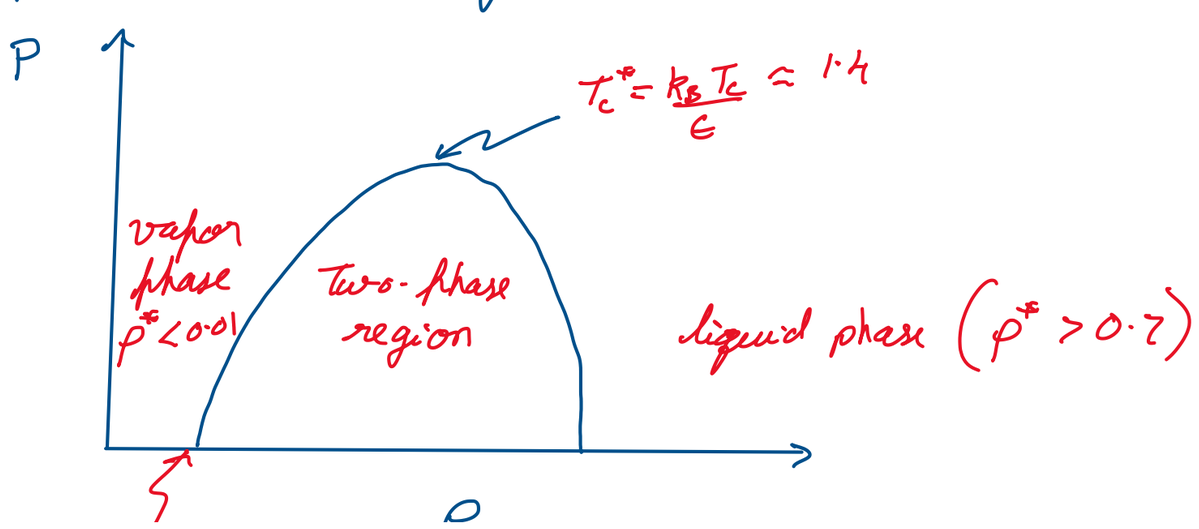
Define Energy scale $u^* = \frac{u}{\epsilon}$

Length scale $r^* = \frac{r}{\sigma}$

$\therefore u^* = 4\left[\left(\frac{1}{r^*}\right)^{12} - \left(\frac{1}{r^*}\right)^{6}\right]$

Property		Reduced Units
Energy	U	$U^* = \frac{U}{\epsilon}$
Length	L	$L^* = \frac{L}{\sigma}$
Pressure	P	$P^* = \frac{P\sigma^3}{\epsilon}$
Number density	ρ	$\rho^* = \rho\sigma^3$
Temperature	T	$T^* = \frac{T}{\epsilon/k_B} = \frac{k_B T}{\epsilon}$
Time	t	$t^* = \frac{t}{\sqrt{\frac{m\sigma^2}{\epsilon}}}$
Surface tension	γ	$\gamma^* = \gamma\frac{\sigma^2}{\epsilon}$

VLE for Lennard-Jones fluid in reduced units



Triple point $T^* \simeq 0.7$

Typical value of σ and ϵ

e.g Argon

$\sigma \equiv 3.405\ \text{Å}$

$\frac{\epsilon}{k_B} = 119.8\ K$

temperature $T^* = 1.0 \Rightarrow T = 119.8\ K$

$\rho^* = 1.0 \quad \rho = 1680\ \frac{kg}{m^3}$

$P^* = 1.0 \quad P = 41.9\ MPa$